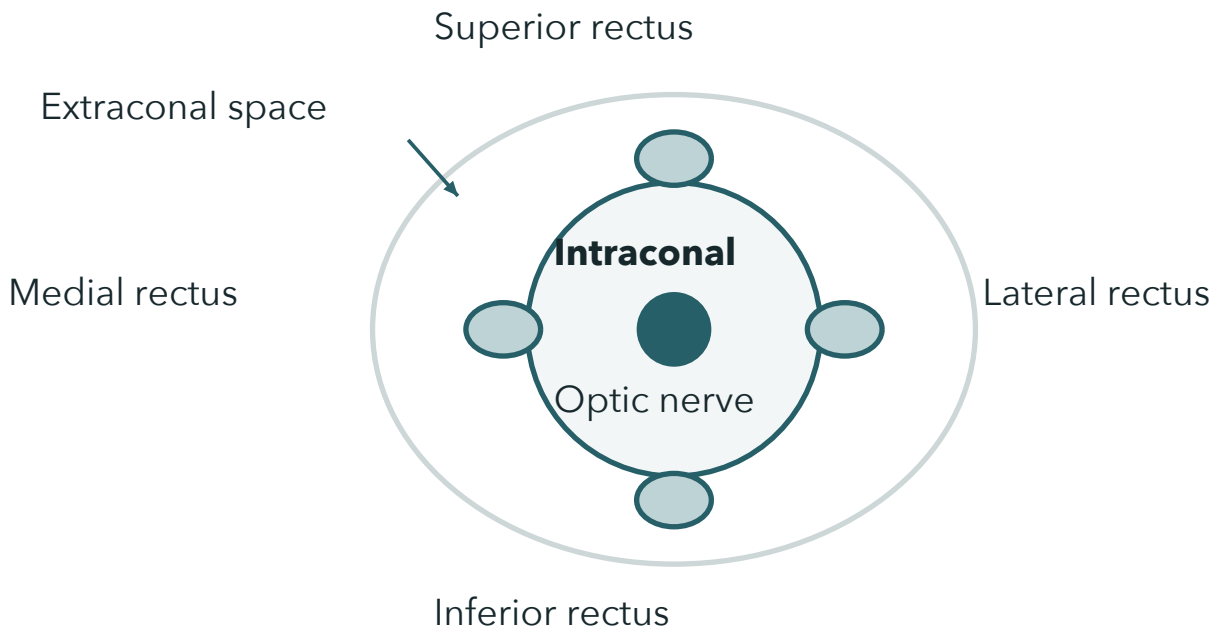


Orbital spaces and the orbital apex

Part 1 | Compartment before nerve name

Localisation is easier when the orbital cone and its apex are separated conceptually.



Coronal schematic: rectus muscles bound the intraconal compartment; optic nerve lies within it.

Spatial rules

Intraconal disease can directly affect optic nerve and ocular motor pathways.

Extraconal disease displaces the cone; displacement alone does not prove invasion.

The orbital septum distinguishes preseptal from postseptal soft tissues.

Apex versus fissure

Afferent visual dysfunction is the key extra clue that the optic-canal region is involved.

Localising pattern

Site	Useful finding
Superior orbital fissure	Motor and V1 deficits; optic nerve spared
Orbital apex	Motor/V1 deficits plus optic neuropathy
Cavernous sinus	May add V2 or sympathetic involvement

Reasoning sequence

- Confirm an afferent visual deficit
- ↓
- Separate retinal from optic-nerve dysfunction
- ↓
- Combine sensory and motor findings
- ↓
- Map the involved openings

Avoid overlocalisation

A relative afferent pupillary defect needs asymmetric afferent dysfunction. Reduced acuity alone can arise from media or macular disease; it is not sufficient evidence that the optic nerve is affected.

Potential spaces and anaesthetic spread

The route of injection predicts the compartment exposed and the possible distant effects.

Plane

Feature	Location	Consequence
Sub-Tenon	Between sclera and Tenon capsule	Spread around globe
Intraconal	Inside rectus cone	Close to optic nerve
Optic-nerve sheath	Meningeal sleeve	Potential central CSF spread

A mechanism, not a diagnosis by timing alone

- Needle enters optic-nerve sheath
- ↓
- Anaesthetic reaches communicating CSF space
- ↓
- Central structures become exposed

Compare alternatives

Intravascular injection and central spread are different mechanisms. An orbital haematoma can cause mechanical optic-nerve compromise but does not itself explain rapid bilateral neurological effects.

Read imaging in three dimensions

An apparent adjacency on one image does not establish a shared tissue plane.

Plane

Feature	View	Question answered
Axial	Medial/lateral relations	Is the cone displaced or invaded?
Coronal	Walls, muscles and nerve	Which compartment is occupied?
Sagittal	Anterior-posterior extent	Does the lesion reach the apex?

Anatomical interpretation

Follow a structure across neighbouring slices.

Distinguish a displaced nerve from one incorporated into a lesion.

Relate the posterior lesion edge to the optic canal and fissure.

Why modality matters

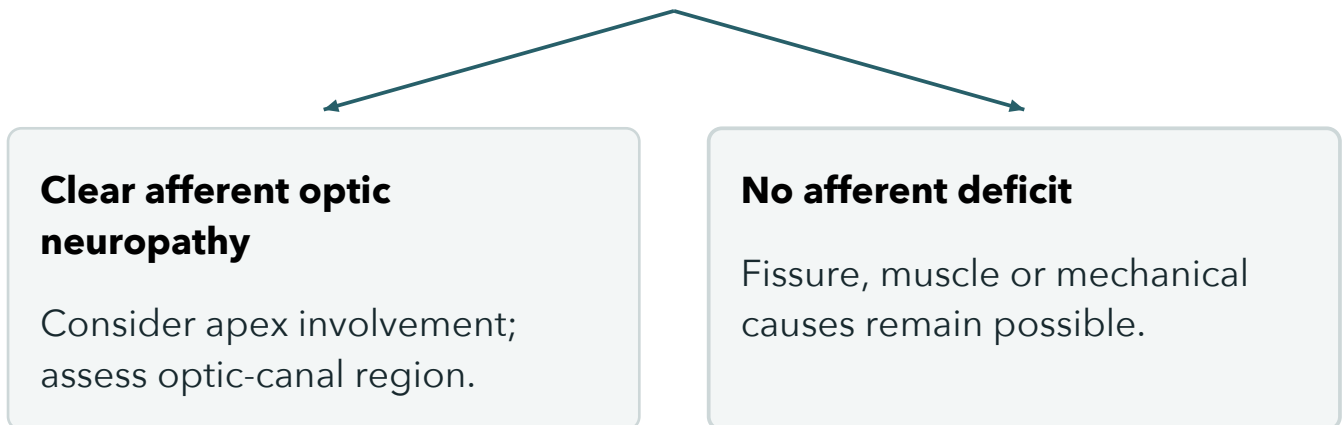
Bone windows reveal wall disruption; soft-tissue imaging characterises the contents. Neither a normal wall nor the word intraconal establishes the biological nature of the lesion.

A localisation rehearsal

Use the full pattern to choose a compartment rather than memorising a list of nerves.

Compare two presentations

What accompanies restricted eye movement?



Three checks before answering

Can a retinal lesion account for the visual findings?

Is the motor deficit neurogenic, restrictive or both?

Does imaging support the proposed compartment?

Postgraduate distinction

The same named nerve can be affected at several sites. A confident localisation requires the combination of affected and spared structures, and a mechanism that explains their anatomical proximity.